IB PHYSICS E.4 FISSION

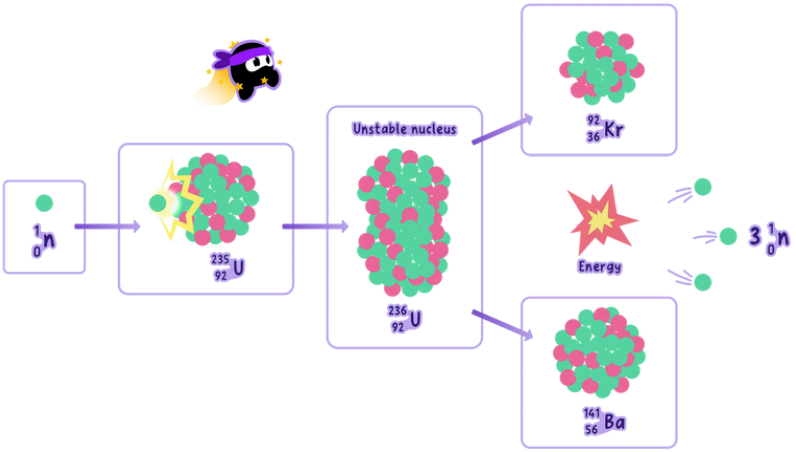
Spontaneous vs induced fission

- **Spontaneous fission**: rare; very heavy nuclei (e.g., U-238, Cf-252) split without external trigger.
- **Induced fission**: neutron absorbed → nucleus unstable → splits into smaller nuclei + neutrons + energy.
- **Energy source**: mass defect → binding energy difference (via E = mc^2).



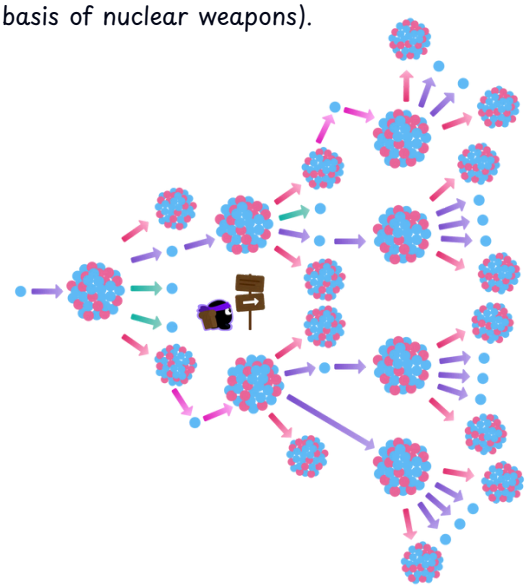
Example:

To the right, you can see an example of induced fission of uranium -235.



Chain reactions and critical mass

- **Chain reaction**: neutrons from one fission trigger further fission events.
- **Critical mass**: minimum fissile material needed for self-sustaining chain reaction.
- **Below critical mass** → neutrons escape, reaction stops.
- **Above critical mass** → uncontrolled energy release (basis of nuclear weapons).



Energy release in fission

- **Mass defect**: products' total mass < original mass.
- **Energy released** = Δmc^2.
- **Binding energy curve**: heavy nuclei (U) → split into medium nuclei (Ba, Kr) with higher binding energy per nucleon.
- Difference in binding energies → large energy release (~200 MeV per fission).

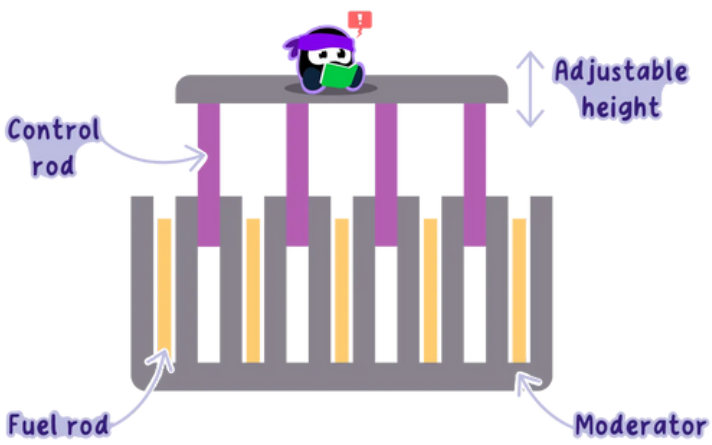


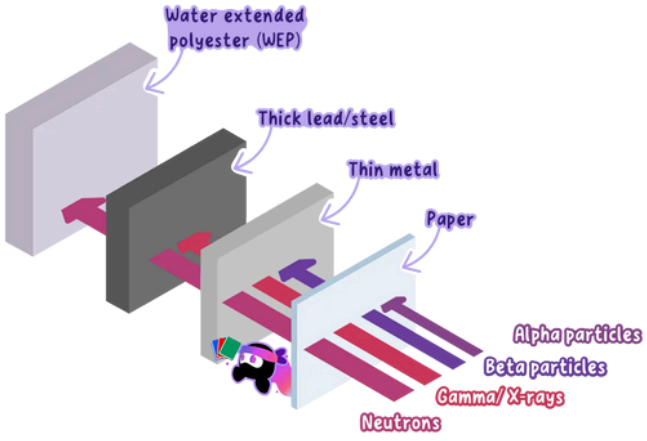
Note!

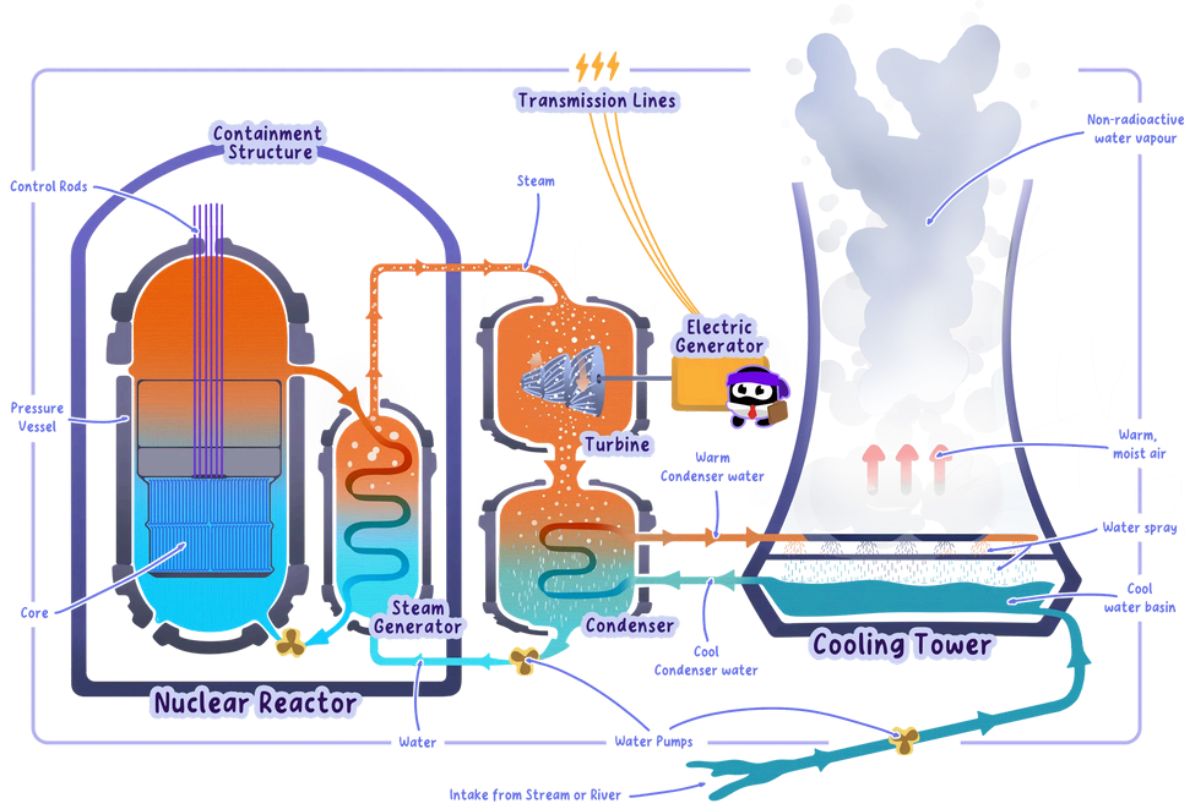
Iron (A≈56) is most stable → all nuclear reactions move curve toward iron.

Components of a nuclear reactor

- **Fuel rods**: contain fissile U-235 or Pu-239.
- **Moderators**: slow neutrons → increase chance of absorption (water, graphite).
- **Control rods**: absorb neutrons → regulate reaction (boron, cadmium).
- **Heat exchanger**: transfers thermal energy → steam turbines → electricity.
- **Shielding**: thick concrete/lead → protects from radiation.







Properties and management of fission products

- **Fission products**: medium-mass isotopes (e.g., Kr-92, Cs-137, Sr-90).
- Usually **radioactive** → emit β, γ radiation.
- **High activity**, varied half-lives (seconds → thousands of years).
- **Management strategies**:
  - Short term: store in **water pools** (cooling + radiation shielding).
  - Long term: move to **dry casks** or **deep geological storage**.
  - Reprocessing: recover usable isotopes, reduce waste

Note!

High-level waste (HLW) requires isolation for thousands of years.

Common Question

- Explain the roles of moderators and control rods in a nuclear reactor.
- Outline how fission products are stored and managed safely.